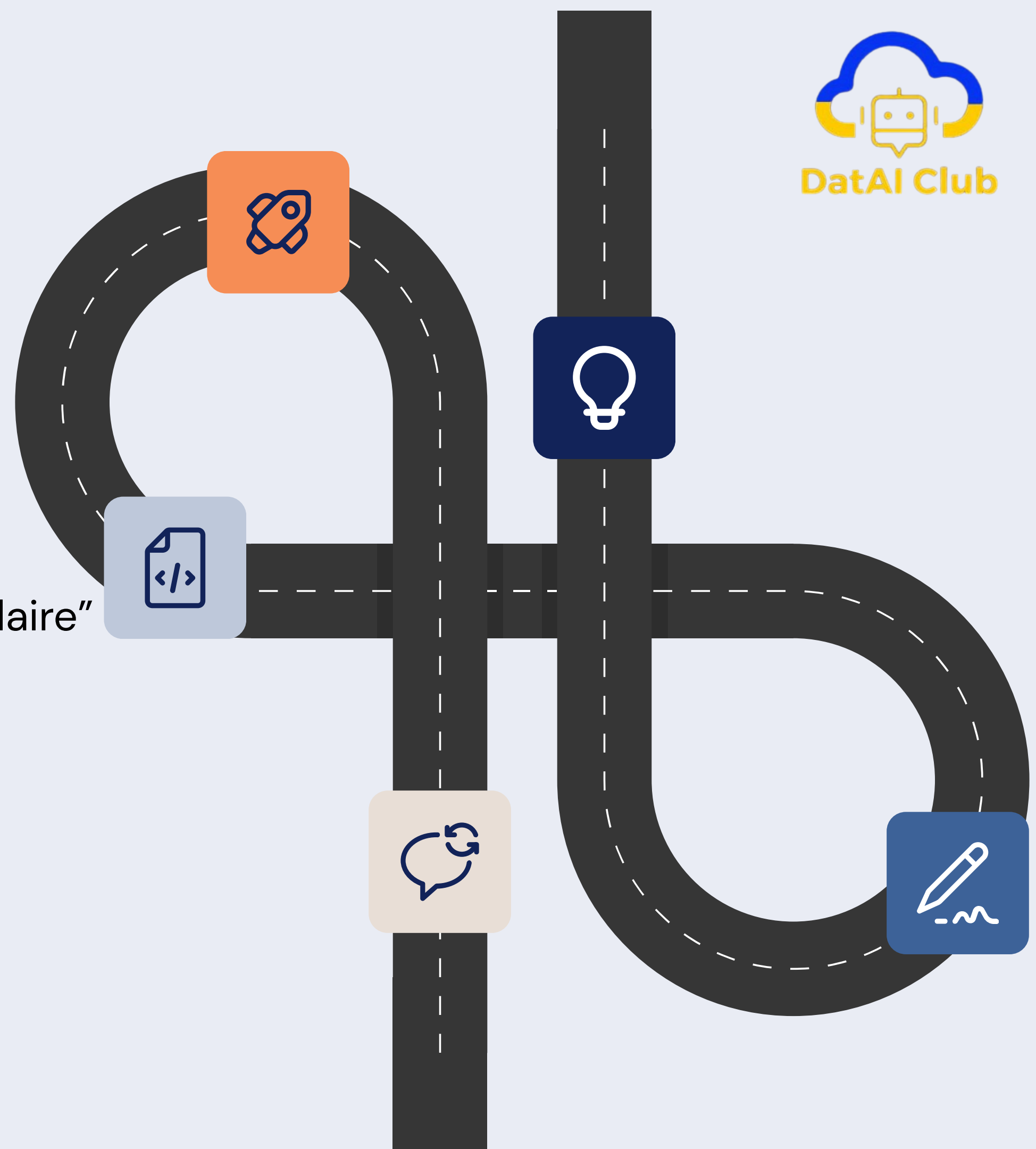
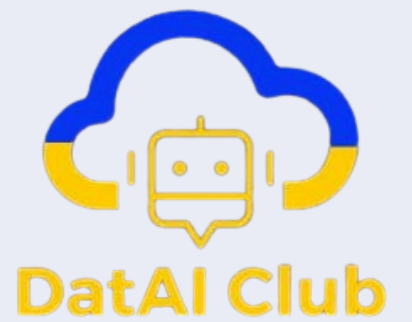


Analyse des Performances des Étudiants

"Étude sur les facteurs influençant la réussite scolaire"

Réalisé par:

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Introduction & Objectifs

Dataset Source : Kaggle - Students Performance in Exams

Population : 1 000 étudiants

01

**Distribution par
Genre**

02

**Taux de Réussite
par Matière**

03

**Impact de
l'Éducation Parentale**

04

**Influence de la
Préparation aux
Examens**

Aperçu des Données

Bibliothèques & Structure du Dataset

Bibliothèques Utilisées



pandas

Manipulation et analyse des données



matplotlib

Visualisations graphiques



seaborn

Visualisations statistiques avancées

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("StudentsPerformance.csv")
print(df.head())
```



Extrait du DataFrame

Gender	Race	Education	Math	Reading	Writing
female	group B	bachelor's	72	72	74
female	group C	some college	69	90	88
female	group B	master's	90	95	93
male	group A	associate's	47	57	44
male	group C	some college	76	78	75

1000

Étudiants

8

Variables

0

Valeurs
manquantes

Distribution par Genre

Analyse de la répartition filles-garçons dans l'échantillon

Code Python

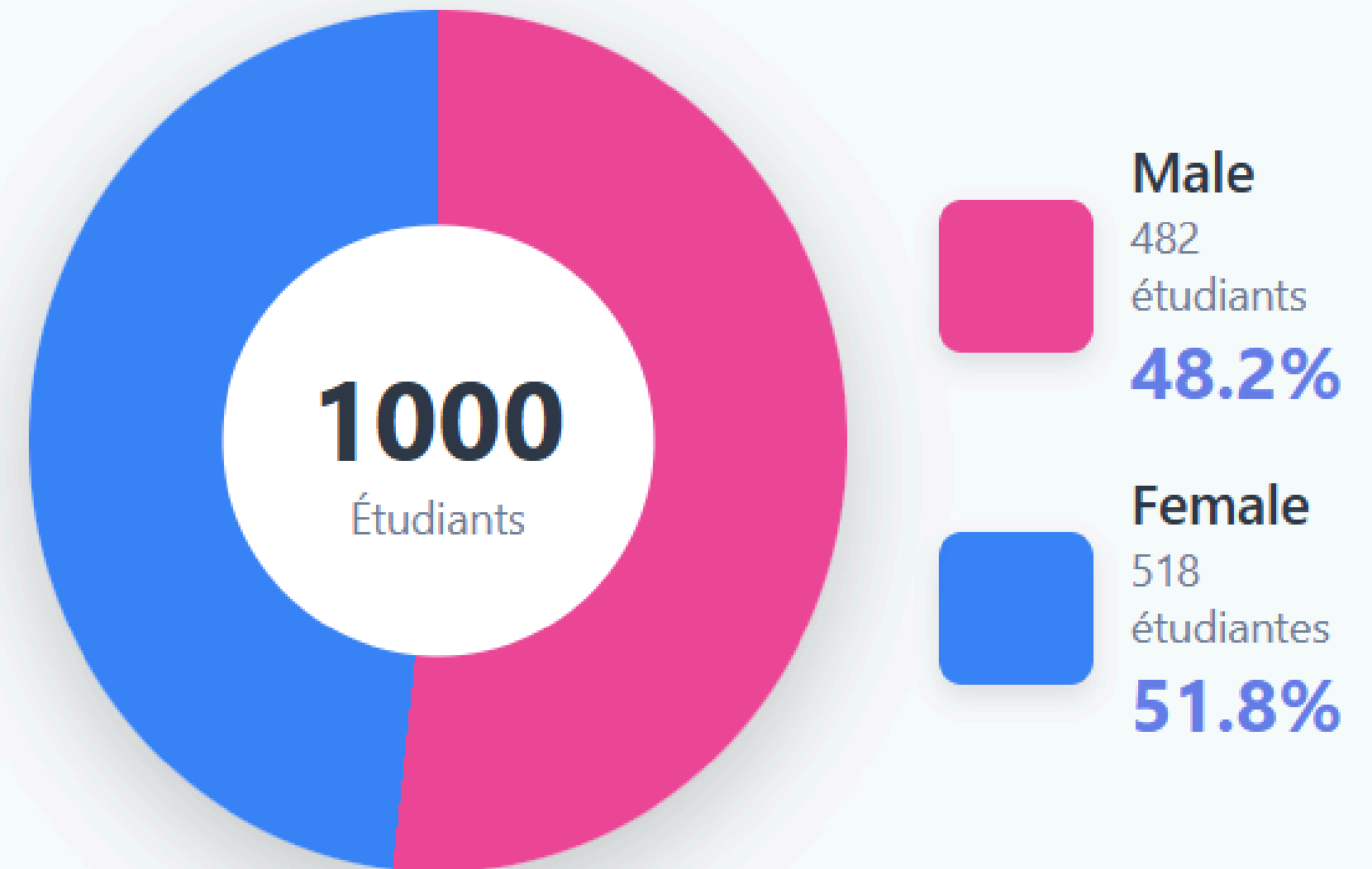
```
print("=== Gender Distribution ===")
print(df['gender'].value_counts())

print("\nPercentage:")
print(df['gender'].value_counts(normalize=True) * 100)
```

```
=== Gender Distribution ===
gender
female    518
male      482
Name: count, dtype: int64

Percentage:
gender
female    51.8
male      48.2
Name: proportion, dtype: float64
```

Visualisation



Réussite en Mathématiques

Analyse du taux de réussite selon le seuil de passage

```
print("\n=== Math Exam Pass/Fail ===")
passed_math = df[df['math score'] >= 40].shape[0]
failed_math = df[df['math score'] < 40].shape[0]

print("Passed Math:", passed_math)
print("Failed Math:", failed_math)

def classify(score):
    if score >= 80: return 'A'
    elif score >= 70: return 'B'
    elif score >= 60: return 'C'
    elif score >= 50: return 'D'
    elif score >= 40: return 'E'
    else: return 'F'

df['math_grade'] = df['math score'].apply(classify)

# Count per grade:
grade_counts = df['math_grade'].value_counts().sort_index()
grade_percent = (df['math_grade'].value_counts(normalize=True) * 100).sort_index()

print("\n=== Grade Distribution (A-F) ===")
print(grade_counts)

print("\n=== Grade Percentages ===")
print(grade_percent)
```

```
=== Math Exam Pass/Fail ===
Passed Math: 960
Failed Math: 40

=== Grade Distribution (A-F) ===
math_grade
A    193
B    216
C    268
D    188
E     95
F     40
Name: count, dtype: int64

=== Grade Percentages ===
math_grade
A    19.3
B    21.6
C    26.8
D    18.8
E     9.5
F     4.0
```

⚙️ Critère de Réussite

Seuil de passage

≥ 40

points sur 100

✓ Réussi

✗ Échoué



960

Étudiants
réussis

96.0%



40

Étudiants
échoués

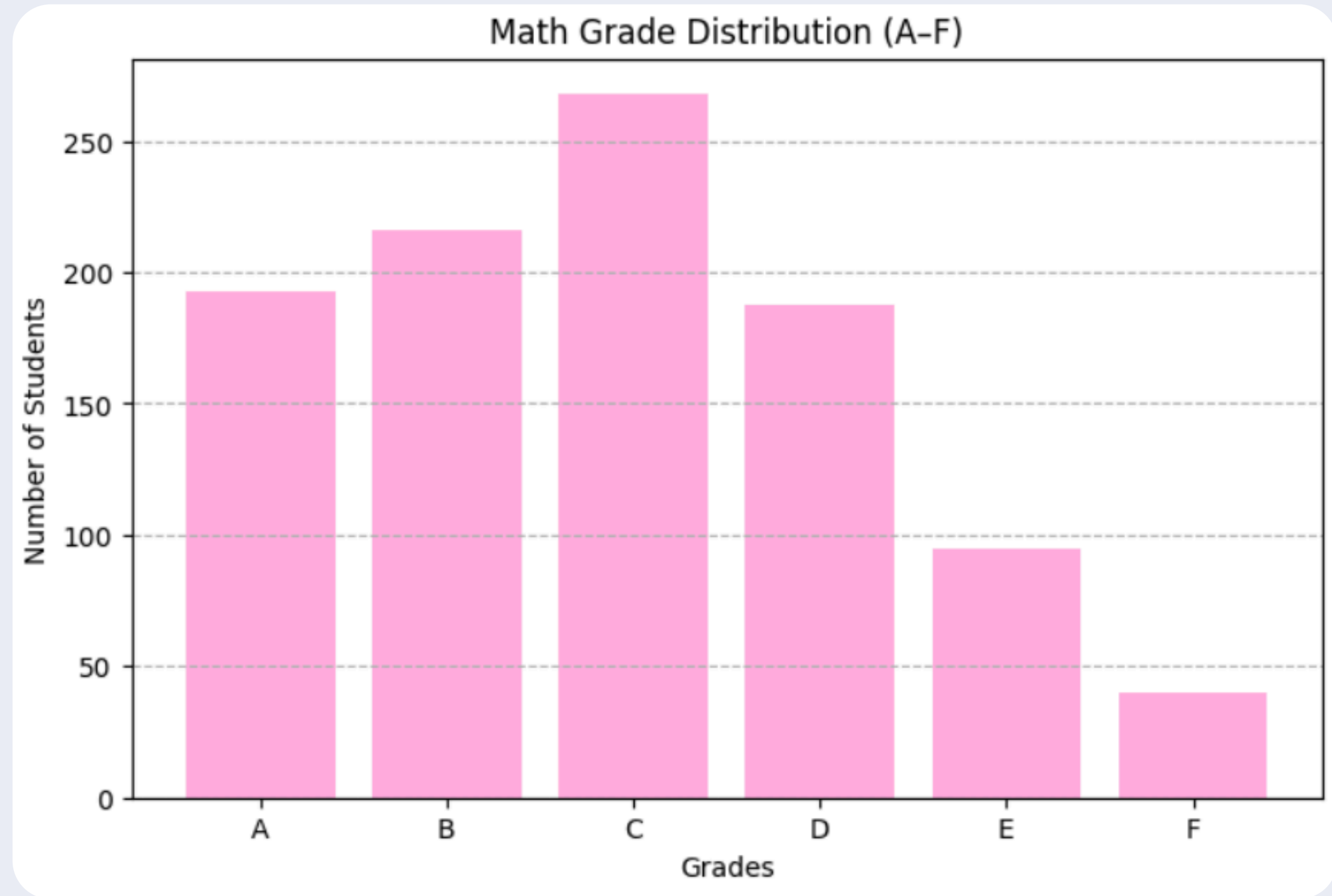
4.0%

Diagramme Math Score

```
plt.figure(figsize=(8,5))
plt.bar(grade_counts.index, grade_counts.values,color="pink")

plt.title("Math Grade Distribution (A-F)")
plt.xlabel("Grades")
plt.ylabel("Number of Students")

plt.grid(axis='y', linestyle='--', alpha=1)
plt.show()
```



Éducation Parentale & Réussite en Maths

Impact du niveau d'éducation des parents sur la performance en mathématiques

Code d'Analyse

```
print("\n=== Parental Education for Math Pass/Fail ===")
print("\nPass:")
print(df[df['math score'] >= 40]['parental level of education'].value_counts())

print("\nFail:")
print(df[df['math score'] < 40]['parental level of education'].value_counts())
```

Affichage des Résultats

```
=== Parental Education for Math Pass/Fail ===

Pass:
parental level of education
some college            220
associate's degree      219
high school             182
some high school        166
bachelor's degree       114
master's degree         59
Name: count, dtype: int64

Fail:
parental level of education
high school             14
some high school        13
some college            6
bachelor's degree       4
associate's degree       3
```

Réussite en Lecture

Analyse du taux de réussite en lecture (seuil ≥ 40 points)

Code Python

```
print("\n=== Reading Exam Pass/Fail ===")
passed_reading = df[df['reading score'] >= 40].shape[0]
failed_reading = df[df['reading score'] < 40].shape[0]

print("Passed Reading:", passed_reading)
print("Failed Reading:", failed_reading)

df['reading_grade'] = df['reading score'].apply(classify)

# Count per grade
grade_countss = df['reading_grade'].value_counts().sort_index()
grade_percentt = (df['reading_grade'].value_counts(normalize=True) * 100).sort_index()

print("\n=== Grade Distribution (A-F) ===")
print(grade_countss)

print("\n=== Grade Percentages ===")
print(grade_percentt)
```

Distribution des Résultats

```
Passed Reading: 974
Failed Reading: 26

=== Grade Distribution (A-F) ===
reading_grade
A      249
B      264
C      233
D      164
E       64
F       26
Name: count, dtype: int64

=== Grade Percentages ===
reading_grade
A      24.9
B      26.4
C      23.3
D      16.4
E       6.4
F       2.6
```


Diagramme Reading Score

```
plt.figure(figsize=(8,5))
plt.bar(grade_countss.index, grade_countss.values,color="purple")

plt.title("Reading Grade Distribution (A-F)")
plt.xlabel("Grades")
plt.ylabel("Number of Students")

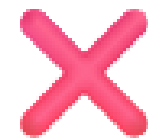
plt.grid(axis='y', linestyle='--', alpha=1)
plt.show()
```



974

Étudiants
réussis

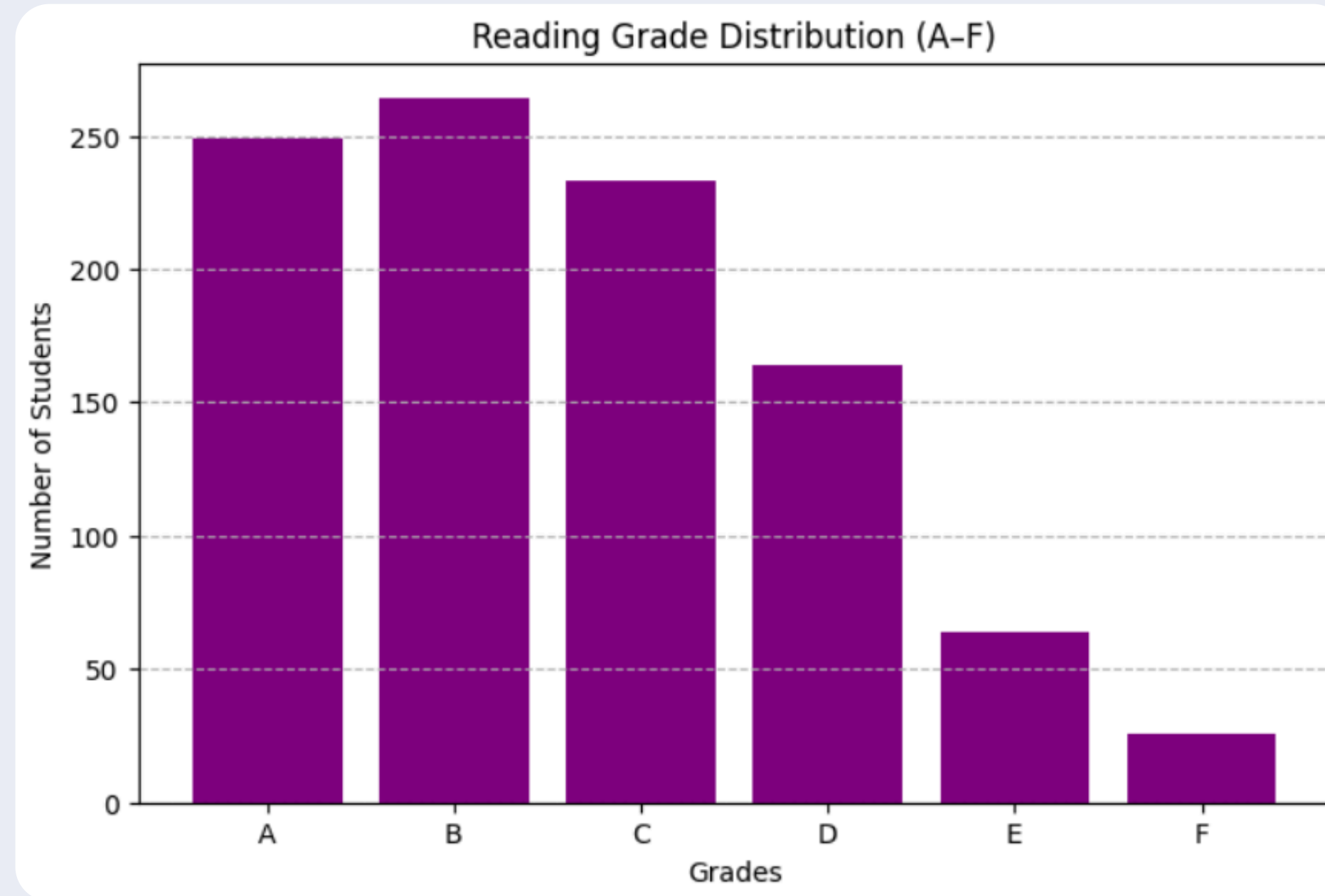
97.4%



26

Étudiants
échoués

2.6%



Éducation Parentale & Réussite en Lecture

Impact du niveau d'éducation des parents sur la performance en lecture

Code d'Analyse

```
print("\n=== Parental Education for Reading Pass/Fail ===")
print("\nPass:")
print(df[df['reading score'] >= 40]['parental level of education'].value_counts())

print("\nFail:")
print(df[df['reading score'] < 40]['parental level of education'].value_counts())
```

Affichage des Résultats

```
=== Parental Education for Reading Pass/Fail ===

Pass:
parental level of education
associate's degree      220
some college            219
high school             189
some high school        169
bachelor's degree       118
master's degree         59
Name: count, dtype: int64

Fail:
parental level of education
some high school        10
high school              7
some college             7
associate's degree       2
```

Réussite Globale (Toutes Matières)

Analyse des étudiants réussissant toutes les matières (Maths, Lecture, Écriture)

Critère de Réussite

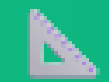
```
print("\n=== Students Who Passed ALL Subjects (>=40) ===")
passed_all = df[(df['math score'] >= 40) &
                (df['reading score'] >= 40) &
                (df['writing score'] >= 40)]

print("Total passed all subjects:", passed_all.shape[0])

print("\nParental level of education for passed-all:")
print(passed_all['parental level of education'].value_counts())

print("\n=== Students Who Failed ALL Subjects (<40) ===")
failed_all = df[(df['math score'] < 40) &
                (df['reading score'] < 40) &
                (df['writing score'] < 40)]

print("Total failed all subjects:", failed_all.shape[0])
print("\nParental level of education for failed-all:")
print(failed_all['parental level of education'].value_counts())
```



Maths

960 (96.0%)



Lecture

974 (97.4%)



Écriture

982 (98.2%)



TOUTES

949 (94.9%)

Affichage des Résultats

```
=== Students Who Passed ALL Subjects (>=40) ===  
Total passed all subjects: 949
```

```
Parental level of education for passed-all:  
parental level of education
```

```
associate's degree    217  
some college          216  
high school           180  
some high school      163  
bachelor's degree     114  
master's degree       59  
Name: count, dtype: int64
```

```
=== Students Who Failed ALL Subjects (<40) ===  
Total failed all subjects: 18
```

```
Parental level of education for failed-all:  
parental level of education
```

```
some high school      7  
high school           5  
some college          5  
associate's degree    1
```

Résultats Globaux



949

Réussissent tout

94.9%



18

Échouent TOUTES

1.8%

Profil des Top Performers



	Master's	11.4%
	Bachelor's	14.5%
	Associate's	21.2%
	Some college	21.5%
	Some HS	17.4%
	High school	17.4%

La moyenne de chaque étudiant dans les 3 matières

```
df["average_score"] = df[["math score", "reading score", "writing score"]].mean(axis=1)
df['average_grade'] = df['average_score'].apply(classify)
print("\nThe average score of students across all subjects")
print(df[["math score", "reading score", "writing score", "average_score", "average_grade"]])
```

The average score of students across all subjects

	math score	reading score	writing score	average_score	average_grade
0	72	72	74	72.666667	B
1	69	90	88	82.333333	A
2	90	95	93	92.666667	A
3	47	57	44	49.333333	E
4	76	78	75	76.333333	B
..	...	...	...	...	...
995	88	99	95	94.000000	A
996	62	55	55	57.333333	D
997	59	71	65	65.000000	C
998	68	78	77	74.333333	B
999	77	86	86	83.000000	A

Grades & Éducation Parentale

Corrélation entre le niveau d'éducation des parents et la distribution des notes

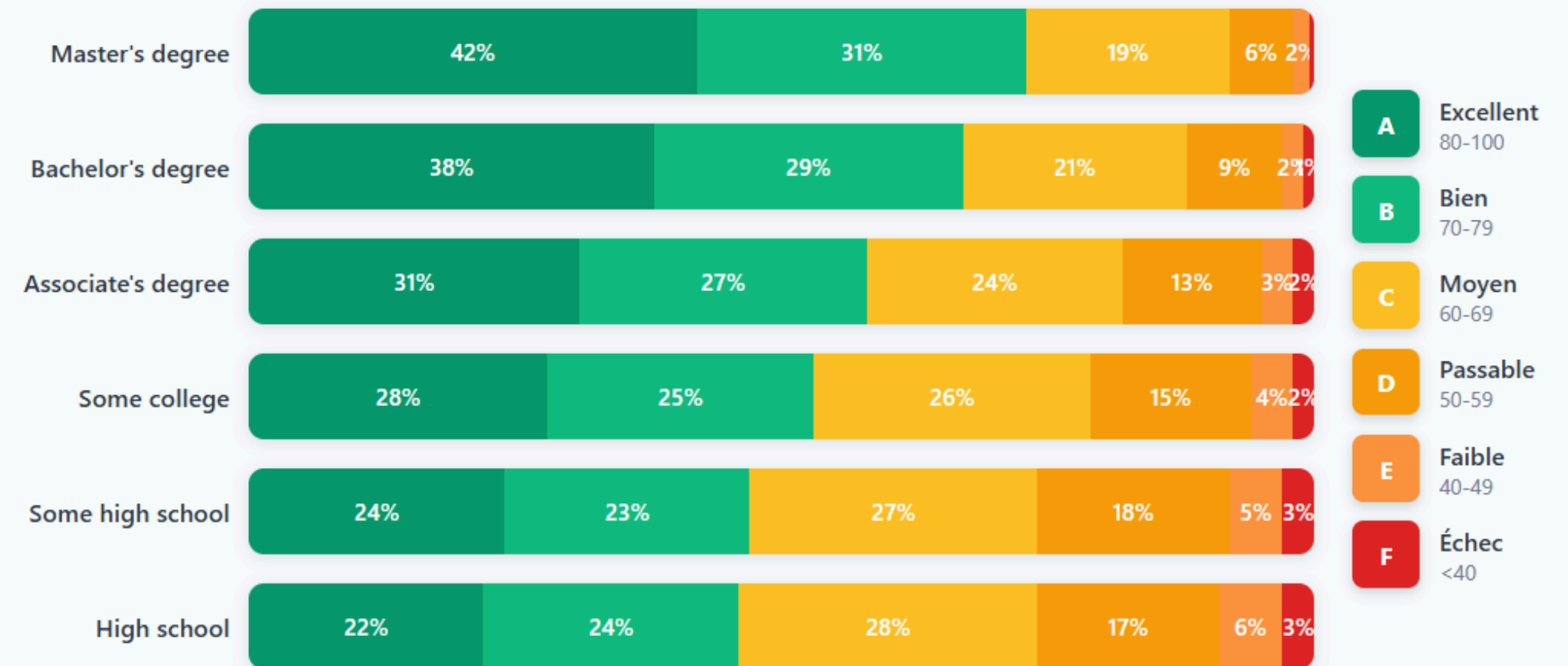
Code d'Analyse

```
print("\n=== Parental Education for Each Grade ===")
grades = ['A', 'B', 'C', 'D', 'E', 'F']

for grade in grades:
    print(f"\nGrade {grade} :")
    print(df[df['Moyenne_grade'] == grade]['parental level of education'].value_counts())
```



Distribution des Grades par Niveau d'Éducation Parentale



Affichage des Résultats

```
=== Parental Education for Each Grade ===
```

```
Grade A :  
parental level of education  
associate's degree    51  
some college          46  
bachelor's degree     31  
some high school      30  
master's degree       20  
high school           20  
Name: count, dtype: int64
```

```
Grade B :  
parental level of education  
some college          62  
associate's degree    59  
high school           45  
some high school      44  
bachelor's degree     35  
master's degree       16  
Name: count, dtype: int64
```

```
Grade C :  
parental level of education  
some college          63  
high school           55  
associate's degree    53  
some high school      41  
bachelor's degree     31  
master's degree       13
```

```
Grade D :  
parental level of education  
high school           43  
associate's degree    42  
some college          40  
some high school      35  
bachelor's degree     13  
master's degree       9  
Name: count, dtype: int64
```

```
Grade E :  
parental level of education  
high school           23  
some high school      19  
associate's degree    15  
some college          9  
bachelor's degree     6  
master's degree       1  
Name: count, dtype: int64
```

```
Grade F :  
parental level of education  
some high school      10  
high school           10  
some college          6  
associate's degree    2  
bachelor's degree     2
```

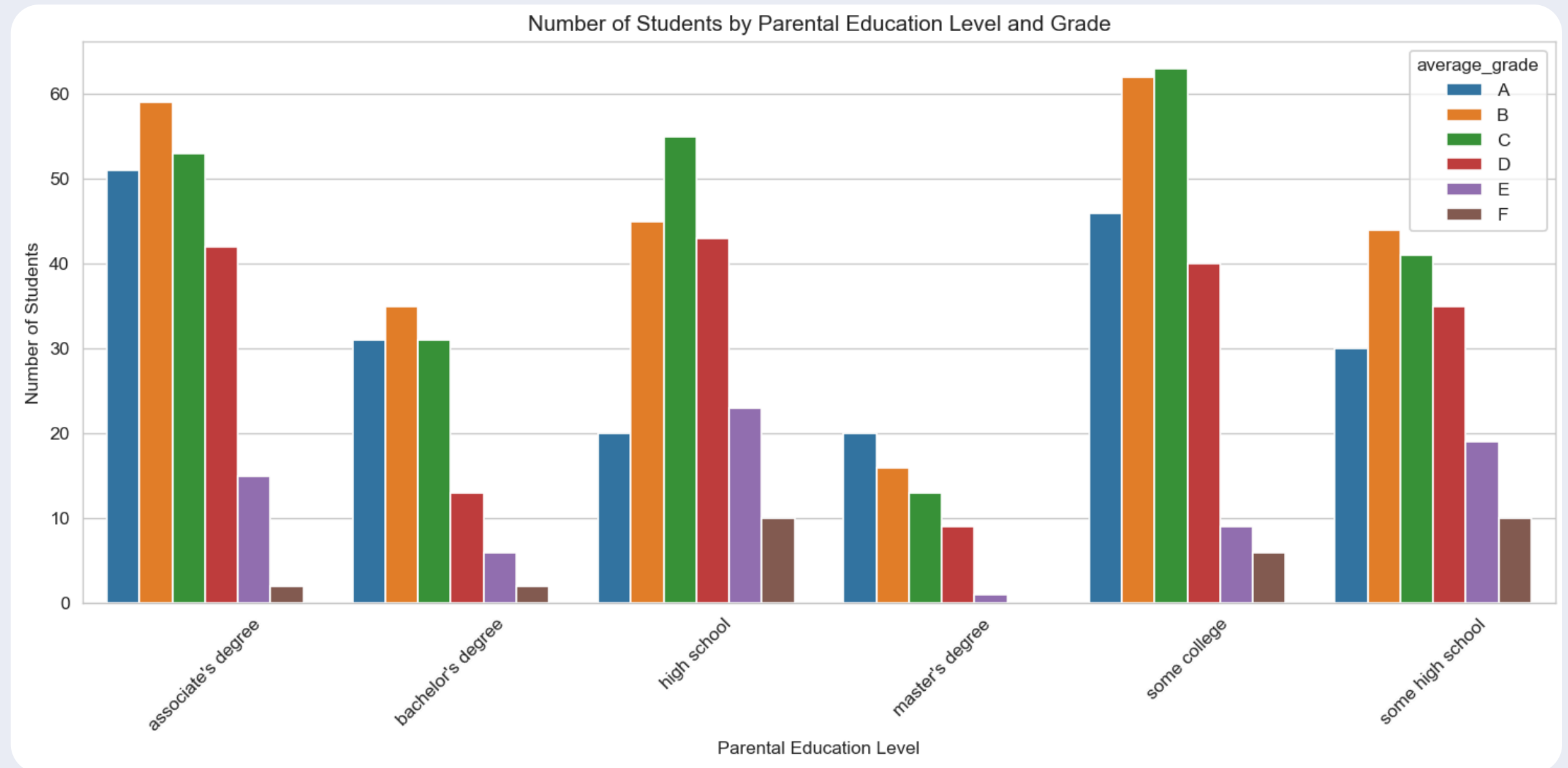
Code Python =>

```
# Créer un DataFrame pour le barplot
count_data = df.groupby(['parental level of education', 'average_grade']).size().reset_index(name='count')

# Plot avec Seaborn
plt.figure(figsize=(12, 6))
sns.set_style("whitegrid")
sns.barplot(
    data=count_data,
    x="parental level of education",
    y="count",
    hue="average_grade"
)

plt.xlabel("Parental Education Level")
plt.ylabel("Number of Students")
plt.title("Number of Students by Parental Education Level and Grade")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Grouped Bar Chart =>



Corrélations entre Matières

Analyse des relations entre les scores de mathématiques, lecture et écriture

Code Python

```
print("\n=== Correlation Matrix ===")  
corr = df[['math score', 'reading score', 'writing score']].corr()  
print(corr)
```

Matrice de Corrélation

	Math	Reading	Writing
Math	1.00 Parfait	0.82 Fort	0.80 Fort
Reading	0.82 Fort	1.00 Parfait	0.95 Très fort
Writing	0.80 Fort	0.95 Très fort	1.00 Parfait

=== Correlation Matrix ===

	math score	reading score	writing score
math score	1.000000	0.817580	0.802642
reading score	0.817580	1.000000	0.954598
writing score	0.802642	0.954598	1.000000

Guide d'Interprétation

- **1.00** - Corrélation parfaite
- **0.90-0.99** - Corrélation très forte
- **0.70-0.89** - Corrélation forte

Impact de la Préparation aux Examens

Comparaison des scores moyens entre étudiants avec et sans préparation

Code Python

```
print("\n=== Test Preparation Course Impact ===")

prep = df.groupby('test preparation course')[['math score', 'reading score', 'writing score']].mean()
print(prepare)
print("\nConclusion:")
print("Students who completed the test preparation tend to have a higher average score, "
      | "while those who did not complete it scored slightly lower on average.")
```

Affichage des Résultats

```
=== Test Preparation Course Impact ===
|  |  |  |  |  |  |  | math score  reading score  writing score
test preparation course
completed          69.695531      73.893855      74.418994
none               64.077882      66.534268      64.504673

Conclusion:
Students who completed the test preparation tend to have a higher average score, while those who did not complete it scored slightly lower on average.
```

Code d'Analyse =>

```
# Making boxplots to relate the tests performance by test preparation
fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(18, 8))

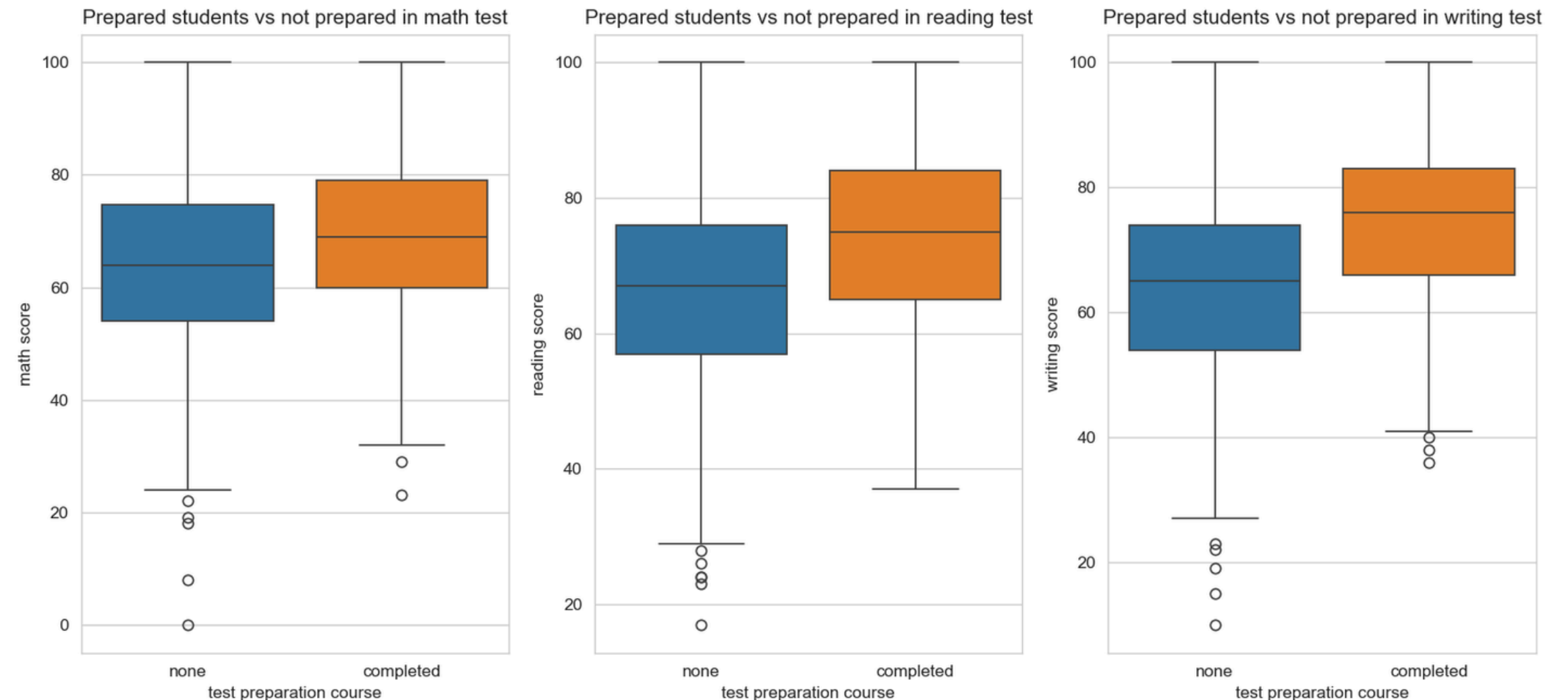
# Create a box plot to show the relationship between completion of preparation course and math score
sns.boxplot(x='test preparation course', y='math score', data=df, hue='test preparation course', ax=axes[0])
axes[0].set_title('Prepared students vs not prepared in math test')

# Create a box plot to show the relationship between completion of preparation course and reading score
sns.boxplot(x='test preparation course', y='reading score', data=df, hue='test preparation course', ax=axes[1])
axes[1].set_title('Prepared students vs not prepared in reading test')

# Create a box plot to show the relationship between completion of preparation course and writing score
sns.boxplot(x='test preparation course', y='writing score', data=df, hue='test preparation course', ax=axes[2])
axes[2].set_title('Prepared students vs not prepared in writing test')

plt.show()
```

Box Plot =>

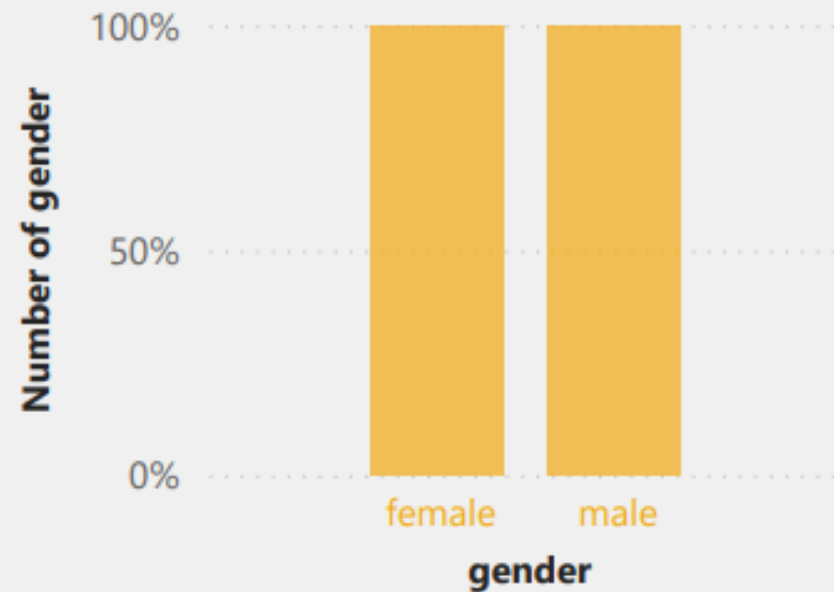




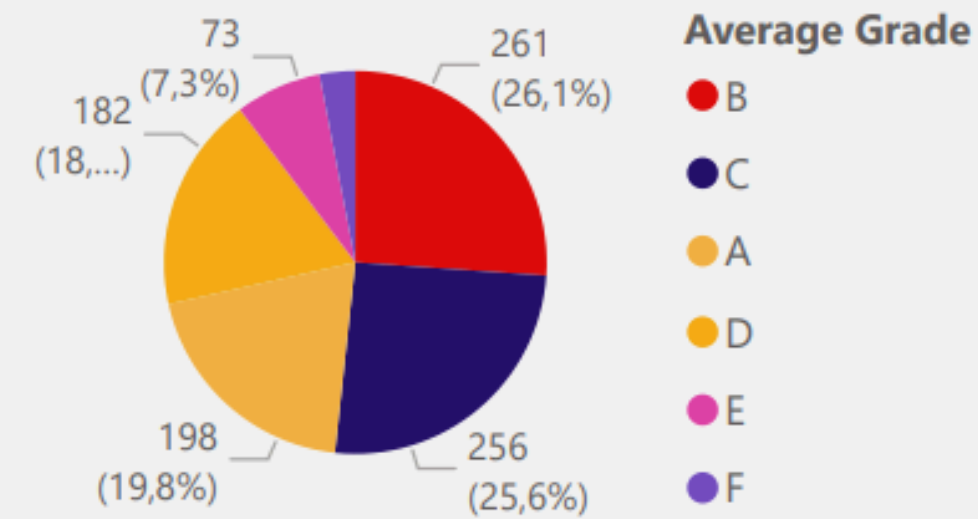
STUDENTS PERFORMANCE ANALYSIS



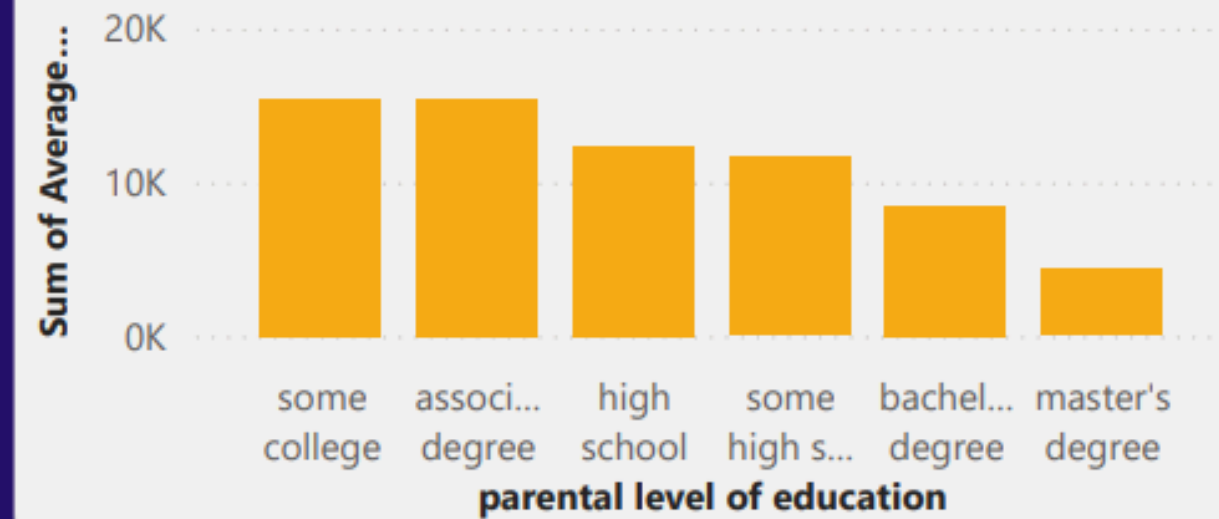
Number of gender by gender



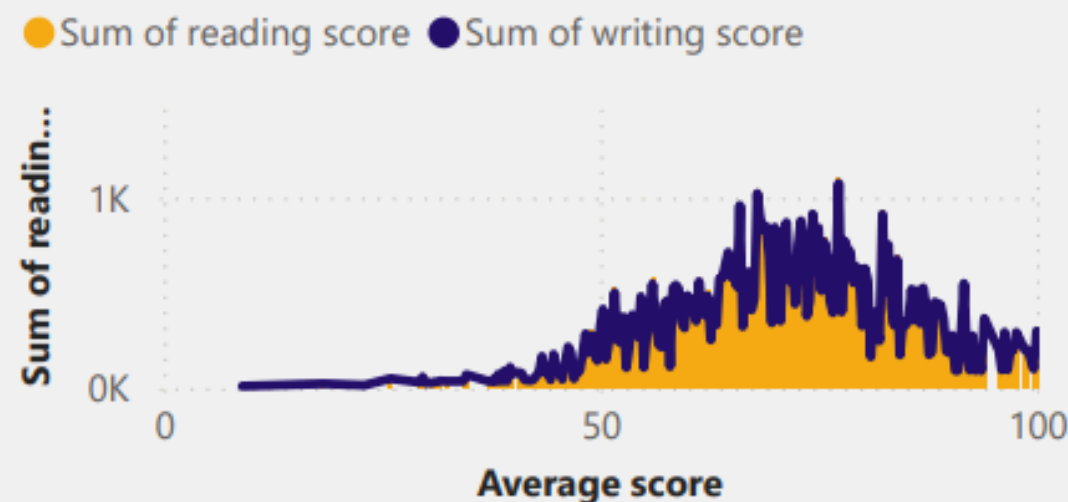
Number of test preparation courses by Average Grade



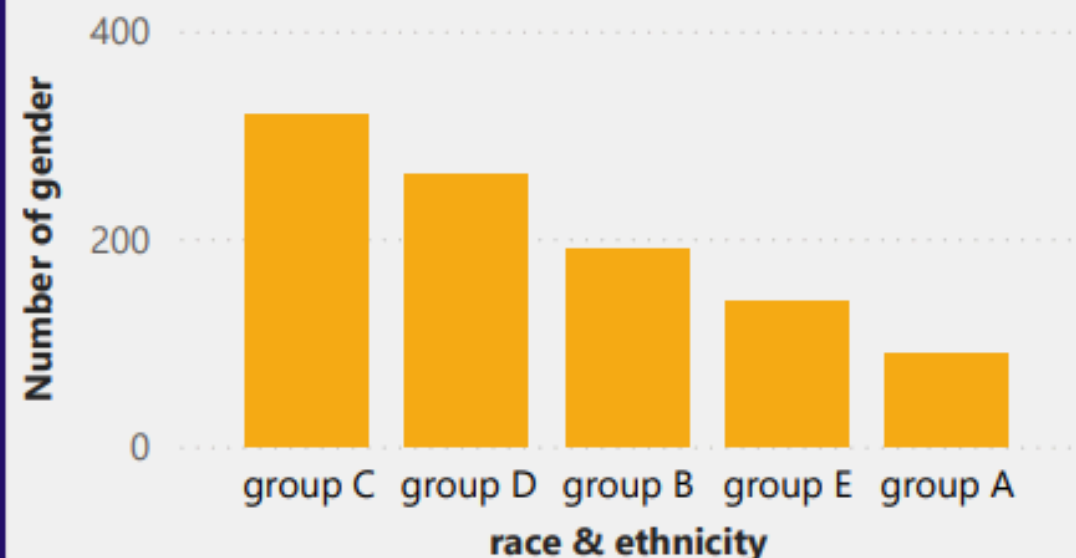
Sum of Average score by parental level of education



Sum of reading score and Sum of writing score by Average score

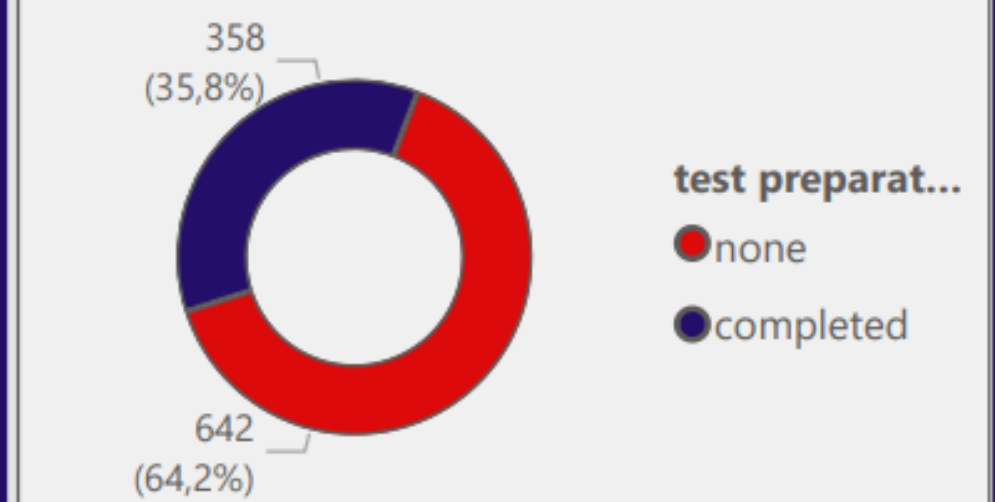


Number of gender by race & ethnicity



Number of Average Grade

by test preparation course





**“ LES DONNÉES RACONTENT
DES HISTOIRES
LES ANALYSTES LES
RENDENT VISIBLES ”**

Merci pour votre Support • **DatAI Club**